

Lab 171: Creación de Instancias Amazon EC2 (Consola y CLI)

- **Dificultad:**  Intermedia
- **Tiempo Estimado:**  45 minutos
- **Servicios Principales:**  **Amazon EC2**,  **AWS CLI**,  **EC2 Instance Connect**,  **Systems Manager (SSM)**.

1. Resumen y Objetivos

Este laboratorio práctico explora las múltiples metodologías para el aprovisionamiento de cómputo en AWS. Implementarás un **Bastion Host** (servidor de salto) mediante la Consola y automatizarás el despliegue de un **Web Server** funcional mediante la interfaz de línea de comandos (CLI). Además, aplicarás técnicas de **Troubleshooting** para corregir fallos de conectividad en entornos mal configurados.

Competencias técnicas:

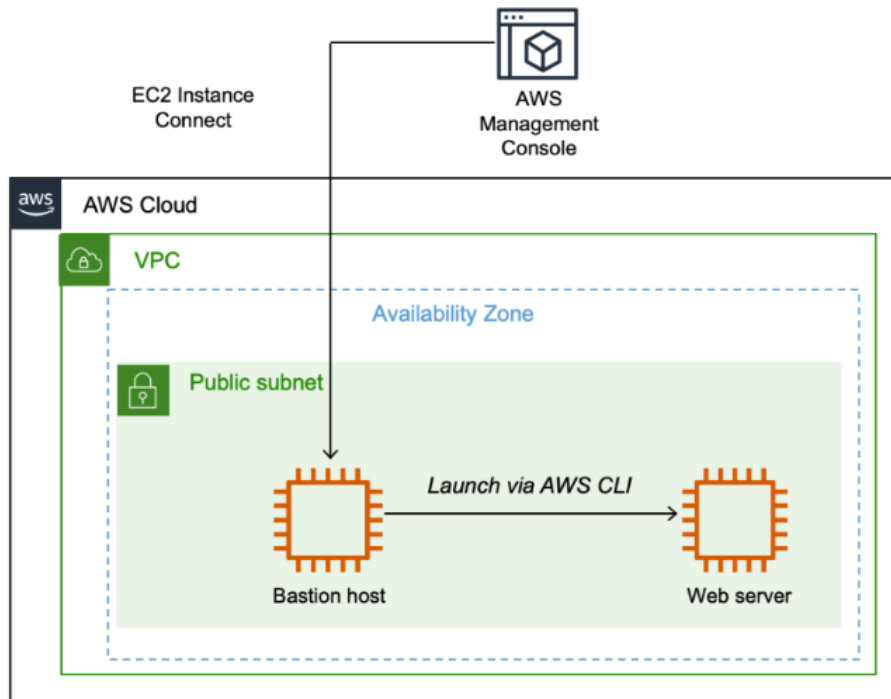
-  Lanzamiento manual de instancias EC2.
-  Conectividad segura mediante EC2 Instance Connect.
-  Automatización programática vía AWS CLI y scripts de **User Data**.
-  Diagnóstico y resolución de problemas de red y seguridad (Capa 3 y 4).

2. Análisis del Escenario

El diseño requiere una arquitectura de gestión segura. Se desplegará un **Bastion Host** en una subred pública para actuar como estación de trabajo administrativa. Desde este punto, el ingeniero debe interactuar con la API de Amazon EC2 para lanzar servidores web de forma consistente y escalable. Finalmente, se debe auditar una instancia preexistente (**Misconfigured Web Server**) para restablecer sus servicios de administración y visualización web.

3. Arquitectura

- **VPC de Laboratorio:** Red segmentada con subredes públicas.
- **Bastion Host:** Estación de administración con un **IAM Role** (**Bastion-Role**) adjunto para permisos de API.
- **Web Server:** Instancia configurada dinámicamente con Apache.
- **Seguridad:** Firewalls virtuales (**Security Groups**) gestionando puertos 22 (SSH) y 80 (HTTP).



4. Desarrollo de las Tareas Paso a Paso

❶ Lanzamiento del Bastion Host (Consola de AWS)

1. **Navega** al servicio **EC2** y haz clic en **Launch instance**.
2. **Configura** los parámetros:
 - **Name:** **Bastion host**.
 - **AMI:** Amazon Linux 2.
 - **Instance type:** **t3.micro**.
 - **Key pair:** Selecciona **Proceed without key pair**.
3. **Configura** los **Network settings**:
 - **VPC:** **Lab VPC**.
 - **Subnet:** **Public Subnet**.
 - **Auto-assign public IP:** **Enable**.
 - **Security group:** Crea uno llamado **Bastion security group** con permiso para **SSH (Puerto 22)**.
4. **Despliega** la sección **Advanced details**:
 - **IAM instance profile:** Selecciona **Bastion-Role**.
5. **Ejecuta** el lanzamiento haciendo clic en **Launch instance**.

Launch an instance | EC2 | us-...
us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#launchInstances

Account ID: 8568-9088-9425
us-east-1-us-east-1-aws-logs-1-6

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Bastion host

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-043ab4148b7bb33e9 (64-bit (x86), uefi-preferred) / ami-00478270b6d87d5d1 (64-bit (x86), uefi)

Free tier eligible

Virtualization: hvm

ENX enabled: true

Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260330.0 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-043ab4148b7bb33e9

Publish Date

2026-03-30

Username

ec2-user

Verified provider

Instance type

Get advice

Instance type

t3.micro

Free tier eligible

Family: t3

2 vCPU

1 GiB Memory

Current generation: true

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0796 USD per Hour

On-Demand Linux base pricing: 0.0704 USD per Hour

All generations

Summary

Number of instances

1

Software image (AMI)

Amazon Linux 2023 AMI 2023.10...read more

ami-043ab4148b7bb33e9

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t3.micro Instance usage (for t3.micro where t3.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

Cancel

Launch instance

Preview code

Launch an instance | EC2 | us-...
us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#launchInstances

Account ID: 8568-9088-9425
us-east-1-us-east-1-aws-logs-1-6

EC2 > Instances > Launch an instance

Instance type

t3.micro 2 vCPU 1 GiB Memory Current generation: true On-Demand SUSE base pricing: 0.0104 USD per Hour Free tier eligible

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour On-Demand Windows base pricing: 0.0796 USD per Hour On-Demand Linux base pricing: 0.0704 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Proceed without a key pair (Not recommended)

Default value

Create new key pair

Network settings

VPC - required

vpc-01767589fe6255887 (Lab VPC)

Subnet

subnet-0b9e5a684022ec0dd

VPC: vpc-01767589fe6255887 Owner: 856890889425 Availability Zone: us-west-2a (usw2-ec2) Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.0.0/24

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

Bastion security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _ (underscore) (0-9, spaces, and _)

Description - required

Permit SSH connections

Inbound Security Group Rules

Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type

ssh

Protocol

TCP

Port range

22

Remove

Summary

Number of instances

1

Software image (AMI)

Amazon Linux 2023 AMI 2023.10...read more

ami-043ab4148b7bb33e9

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

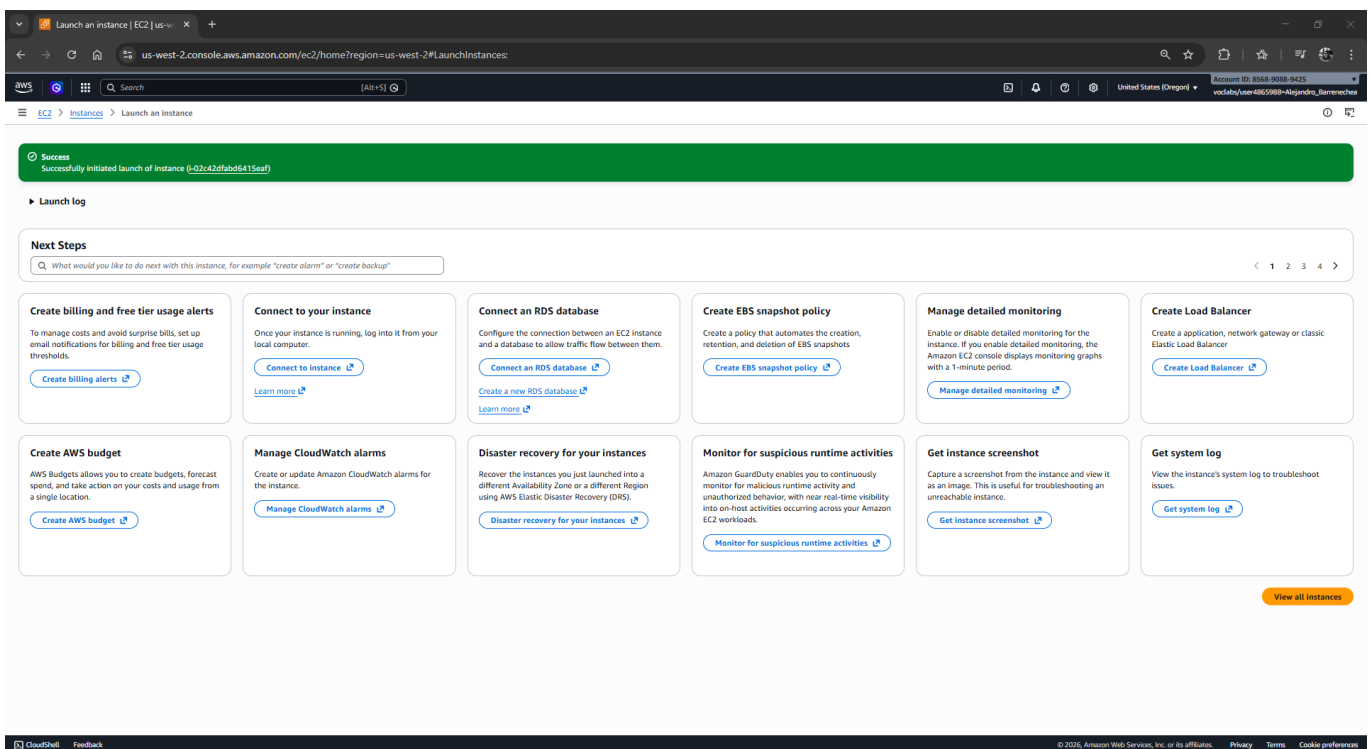
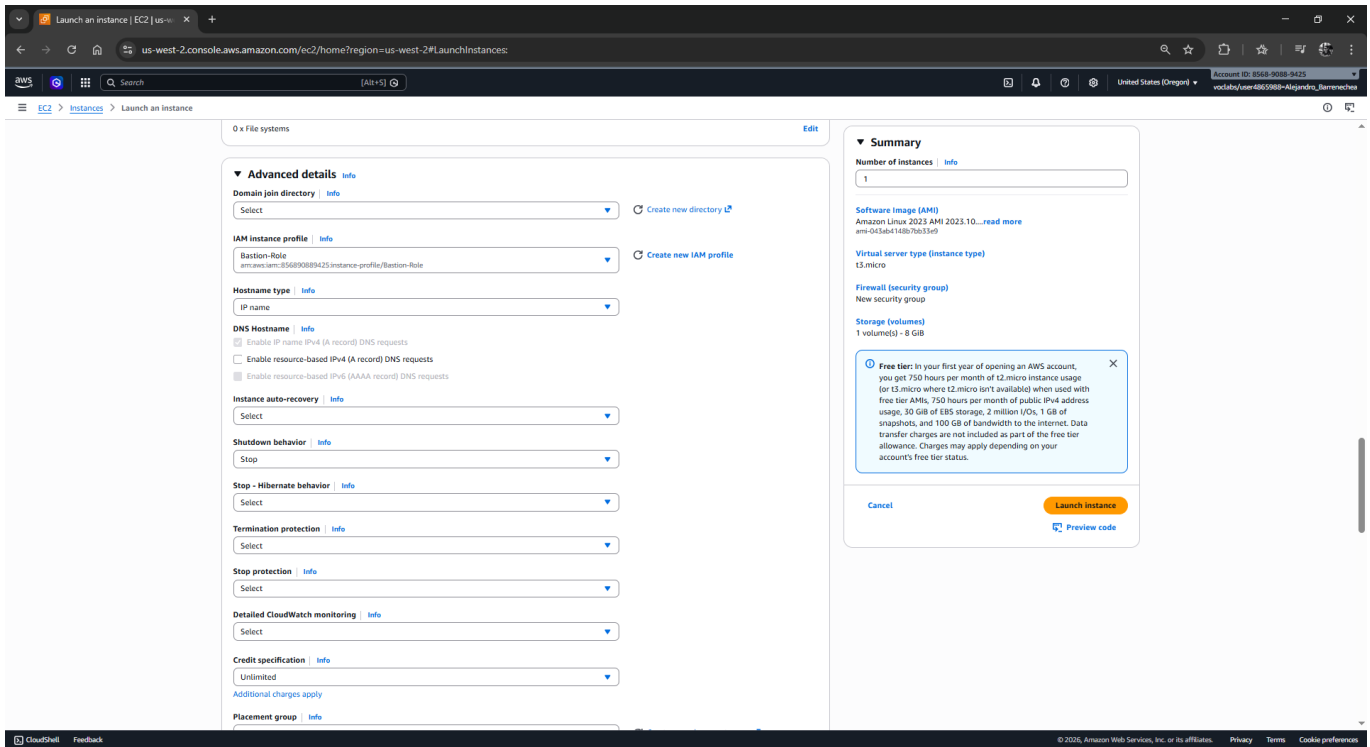
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t3.micro Instance usage (for t3.micro where t3.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

Cancel

Launch instance

Preview code

3 / 14



② Conexión al Host mediante EC2 Instance Connect

1. **Selecciona** tu instancia **Bastion host** en la lista.
2. **Haz clic** en **Connect**.
3. **Usa** la pestaña **EC2 Instance Connect** y presiona **Connect**. Se abrirá una terminal en una nueva pestaña del navegador.

The screenshot shows the AWS Management Console for the 'us-west-2' region. The left sidebar contains navigation links for EC2, including Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, Elastic IPs, Placement Groups, Key Pairs, Network Interfaces, Load Balancing, Load Balancers, Target Groups, Trust Stores, and Auto Scaling, Auto Scaling Groups, and Settings.

The main content area displays the 'Instances' list with two instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security
Bastion host	i-02c42dfabd6415eaf	Running	t3.micro	3/3 checks passed	View alarms	us-west-2a	ec2-54-186-61-65.us-w...	54.186.61.65	-	-	disabled	Bastion s
Misconfigured Web Server	i-0391719c8433c9c5	Running	t2.micro	2/2 checks passed	View alarms	us-west-2a	ec2-18-236-191-26.us...	18.236.191.26	-	-	disabled	Challenge

The 'Bastion host' instance details are shown in the right pane:

- Instance summary**
 - Instance ID: i-02c42dfabd6415eaf
 - IPv4 address: 54.186.61.65 (Public IP)
 - Private IP address: 10.0.0.42
 - Public DNS: ec2-54-186-61-65.us-west-2.compute.amazonaws.com
 - Elastic IP address: -
 - AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations.
 - Auto Scaling group name: -
 - Managed: false
- Instance details**
 - AMI ID: ami-045ab4148b7bb33e9
 - AMI name: al2023-ami-2023.10.20260330.0-kernel-6.1-x86_64
 - Stop protection: Disabled
 - Instance reboot migration: Disabled
 - Monitoring: disabled
 - Allowed image: -
 - Launch time: Thu Apr 02 2025 12:29:48 GMT-0500 (hora de verano de Chile) (3 minutes)
 - Instance auto-recovery: Disabled
 - Platform details: Linux/UNIX
 - Termination protection: Disabled
 - AMI location: amazon/al2023-ami-2023.10.20260330.0-kernel-6.1-x86_64
 - Lifecycle: Standard

The screenshot shows the 'Connect to instance' page for the 'Bastion host' instance. The page provides information about the instance and offers options to connect using a public IP or a private IP.

Instance ID: i-02c42dfabd6415eaf (Bastion host)

VPC ID: vpc-01767589f62f5887

Security groups: sg-0f7a3c46d77f95b53 (Bastion security group)

IAM role: Bastion-Role

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

Instance ID: i-02c42dfabd6415eaf (Bastion host)

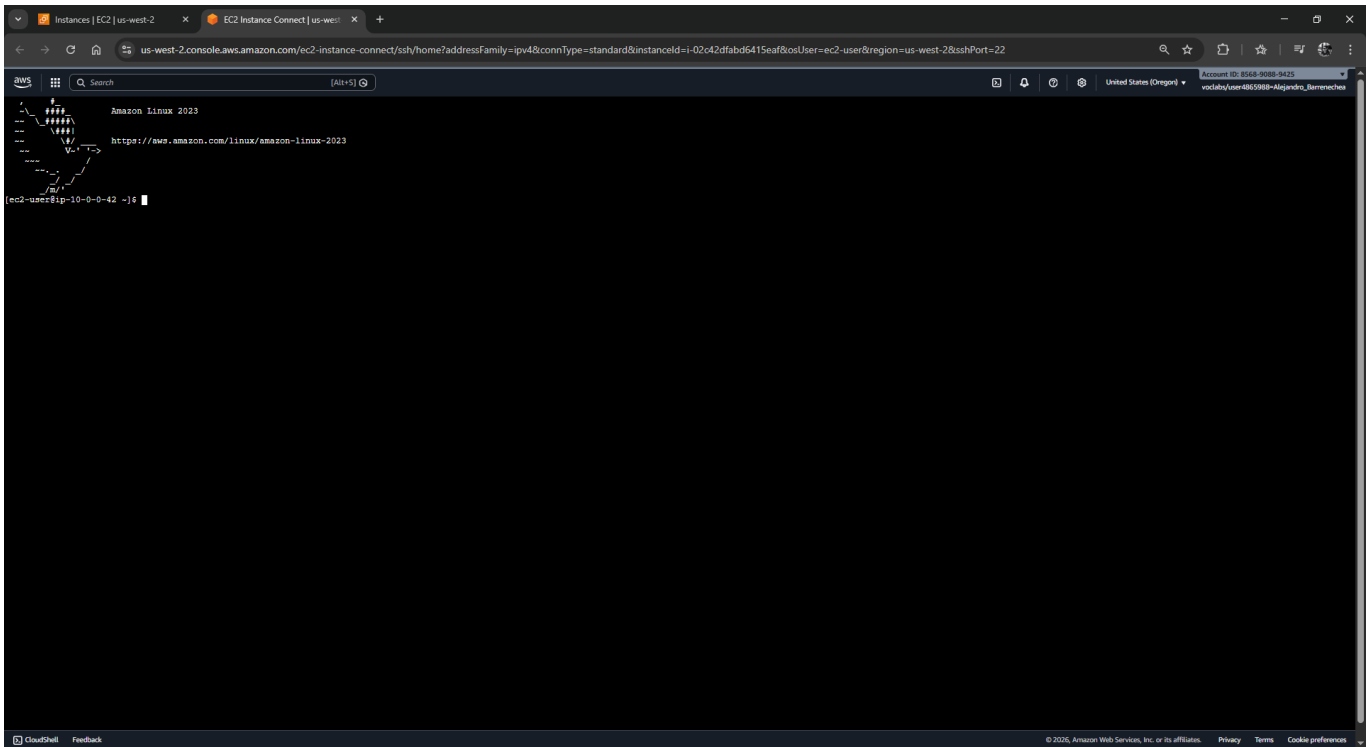
Connection type: ☒ Connect using a Public IP (Public IP or IPv6 address) ☐ Connect using a Private IP (Connect using a private IP address and a VPC endpoint)

☒ Public IPv4 address: 54.186.61.65 ☐ IPv6 address: -

Username: ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect



3 Lanzamiento del Web Server mediante AWS CLI (Automatización)

Dentro de la terminal del Bastion, ejecuta los siguientes comandos técnicos:

1. **Recupera** la AMI más reciente y configura la región:

```
AZ=`curl -s http://169.254.169.254/latest/meta-data/placement/availability-  
zone`  
export AWS_DEFAULT_REGION=${AZ::-1}  
AMI=$(aws ssm get-parameters --names /aws/service/ami-amazon-linux-  
latest/amzn2-ami-hvm-x86_64-gp2 --query 'Parameters[0].[Value]' --output  
text)
```

2. **Identifica** los IDs de red necesarios:

```
SUBNET=$(aws ec2 describe-subnets --filters 'Name=tag:Name,Values=Public  
Subnet' --query Subnets[].SubnetId --output text)  
SG=$(aws ec2 describe-security-groups --filters Name=group-  
name,Values=WebSecurityGroup --query SecurityGroups[].GroupId --output text)
```

```

[ec2-user@ip-10-0-0-42 ~]$ # Obtener Zona de Disponibilidad y Región
az= curl -s http://169.254.169.254/latest/meta-data/placement/availability-zone
export AWS_DEFAULT_REGION=$(az --out text)

# Recuperar la ID de la AMI más reciente desde el Parameter Store
AMI=$(aws ssm get-parameters --names /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2 --query 'Parameters[0].Value' --output text)
echo "La AMI a utilizar es: $AMI"
-bash: -l: substring expression < 0
La AMI a utilizar es: ami-0a37c0e7db254a3

[ec2-user@ip-10-0-0-42 ~]$ # Obtener ID de la Subred Pública
SUBNET=$(aws ec2 describe-subnets --filters 'Name:tag:Name,Values:Public Subnet' --query Subnets[0].SubnetId --output text)

# Obtener ID del Grupo de Seguridad Web
SG=$(aws ec2 describe-security-groups --filters Name=group-name,Values=WebSecurityGroup --query SecurityGroups[0].GroupId --output text)
echo "Subnet ID: $SUBNET | Security Group ID: $SG"
Subnet ID: subnet-0b2a34563022ec0d | Security Group ID: sg-3ec115921ae7ca130
[ec2-user@ip-10-0-0-42 ~]$

```

3. Descarga el script de instalación (User Data):

```
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-RSJAWS-1-23732/171-lab-JAWS-create-ec2/s3/UserData.txt
```

4. Ejecuta el lanzamiento de la instancia:

```

INSTANCE=$(aws ec2 run-instances --image-id $AMI --subnet-id $SUBNET --
security-group-ids $SG --user-data file:///home/ec2-user/UserData.txt --
instance-type t3.micro --tag-specifications 'ResourceType=instance,Tags=
[{'Key=Name,Value=Web Server'}]' --query 'Instances[*].InstanceId' --output
text)

```

5. Verifica el despliegue y obtén la URL:

```

# Espera a que el estado sea "running"
aws ec2 describe-instances --instance-ids $INSTANCE --query
'Reservations[0].Instances[0].State.Name' --output text
# Obtén el DNS público
aws ec2 describe-instances --instance-ids $INSTANCE --query
'Reservations[0].Instances[0].PublicDnsName' --output text

```

```

[ec2-user@ip-10-0-0-42 ~]$ wget https://aws-to-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-R5JMS-1-23732/171-lab-JMS-create-ec2/a3/UserData.txt
--2026-04-02 15:37:13-- https://aws-to-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-R5JMS-1-23732/171-lab-JMS-create-ec2/a3/UserData.txt
Resolving aws-to-largeobjects.s3.us-west-2.amazonaws.com (aws-to-largeobjects.s3.us-west-2.amazonaws.com)... 52.215.184.169, 3.5.85.219, 52.92.180.154, ...
Connecting to aws-to-largeobjects.s3.us-west-2.amazonaws.com (aws-to-largeobjects.s3.us-west-2.amazonaws.com)|52.215.184.169|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 327 [text/plain]
Saving to: 'UserData.txt'

UserData.txt
100%[----->] 327 --.-KB/s 1in 0s

2026-04-02 15:37:13 (12.7 MB/s) - 'UserData.txt' saved [327/327]

[ec2-user@ip-10-0-0-42 ~]$ aws ec2 run-instances \
--image-id $AMI \
--subnet-id $SUBNET \
--security-groups $SGS \
--user-data file:///home/ec2-user/UserData.txt \
--instance-type $t.micro \
--tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=Web Server}]' \
--query 'Instances[*].InstanceId' \
--output text \
do "Instancia creada con ID: $INSTANCE"
Instancia creada con ID: i-06c6fbcd9d96a24b
[ec2-user@ip-10-0-0-42 ~]$

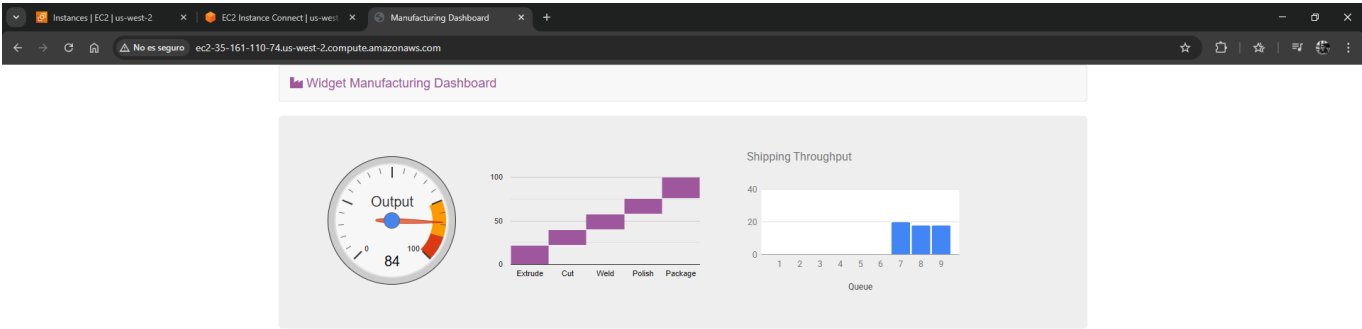
```

```

[ec2-user@ip-10-0-0-42 ~]$ aws ec2 describe-instances --instance-ids $INSTANCE --query 'Reservations[0].Instances[0].State.Name' --output text
running
[ec2-user@ip-10-0-0-42 ~]$ aws ec2 describe-instances --instance-ids $INSTANCE --query 'Reservations[0].Instances[0].PublicDnsName' --output text
ec2-35-161-110-74.us-west-2.compute.amazonaws.com
[ec2-user@ip-10-0-0-42 ~]$ ^C
[ec2-user@ip-10-0-0-42 ~]$

```

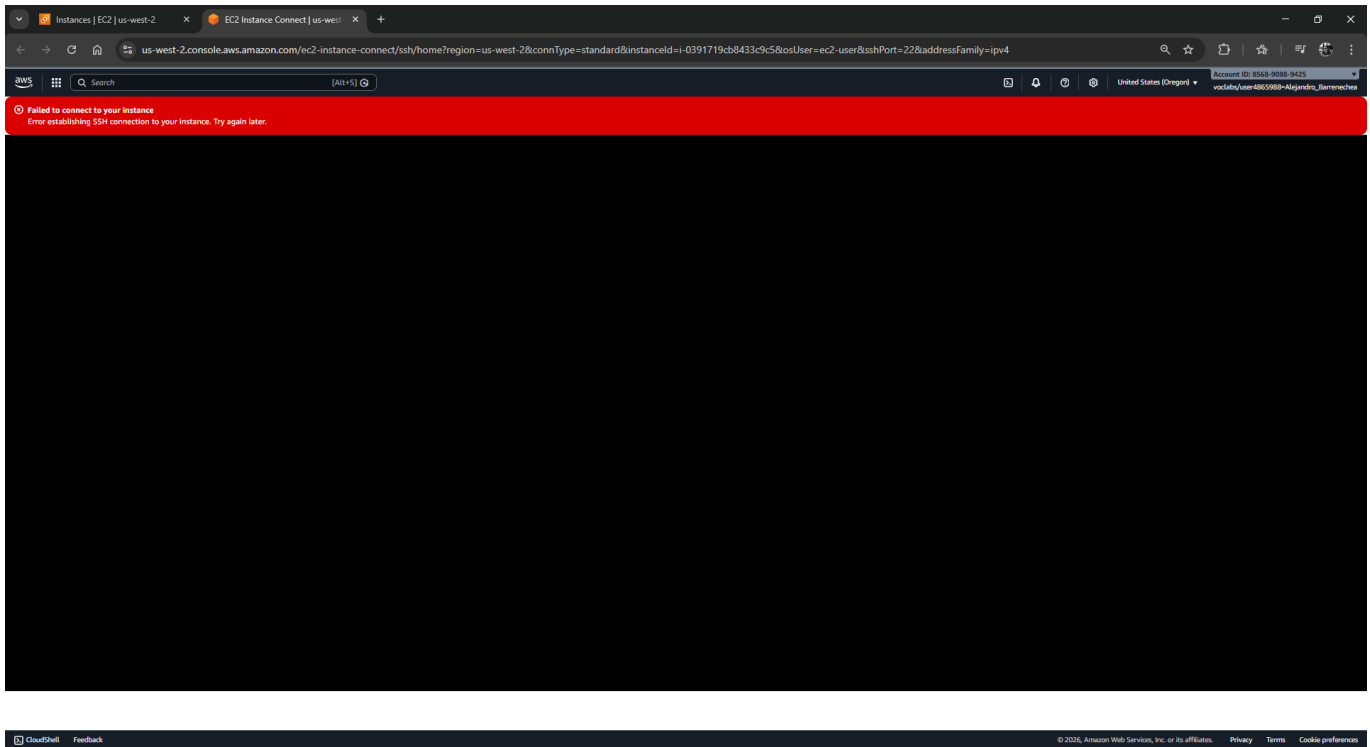
6. **Pega** el DNS en el navegador asegurándote de escribir **http://** al inicio (ej: **http://ec2-xx-xxx.compute.amazonaws.com**).



4 Desafío Opcional 1: Diagnóstico de Conexión (SSH)

- 1. **Tarea:** Intenta conectar a la instancia **Misconfigured Web Server** vía Instance Connect.
- 2. **Fallo detectado:** Error de *timeout* (Puerto 22 bloqueado).

The screenshot shows the 'Connect to instance' page in the AWS console. It displays details for an instance named 'Misconfigured Web Server' (Instance ID: i-0391719cb8433c9c5). The page includes tabs for 'EC2 Instance Connect', 'SSM Session Manager', 'SSH client', and 'EC2 serial console'. Under 'EC2 Instance Connect', the 'Connection type' is set to 'Public IP4 address' (18.236.191.26). The 'Username' is 'ec2-user'. A note at the bottom states: 'Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.'

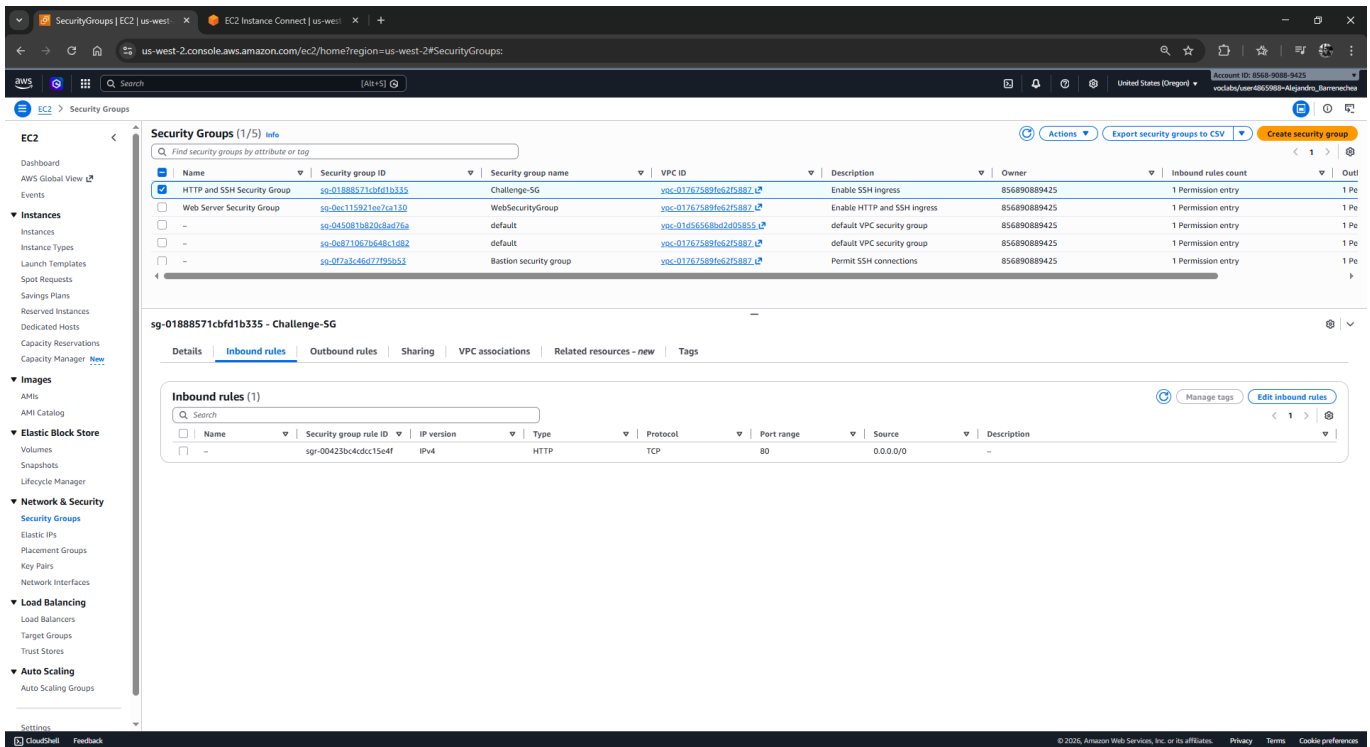


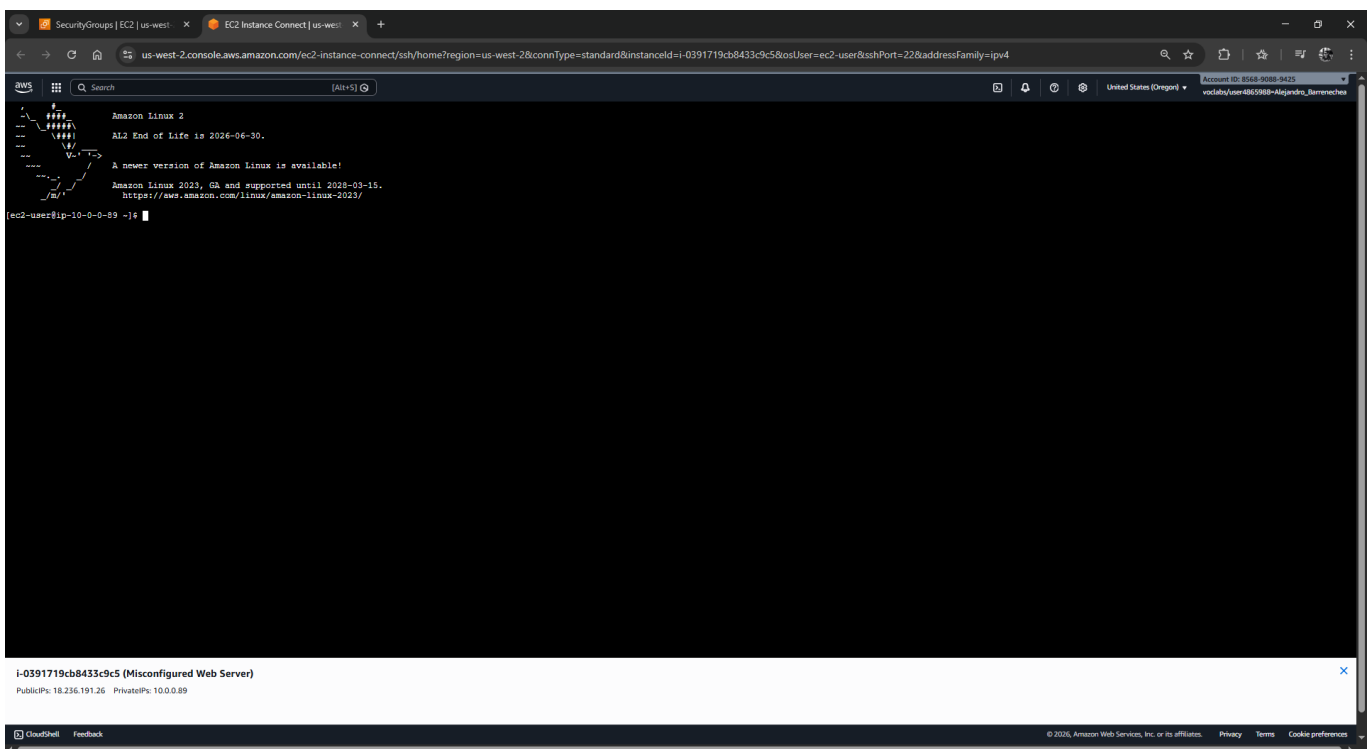
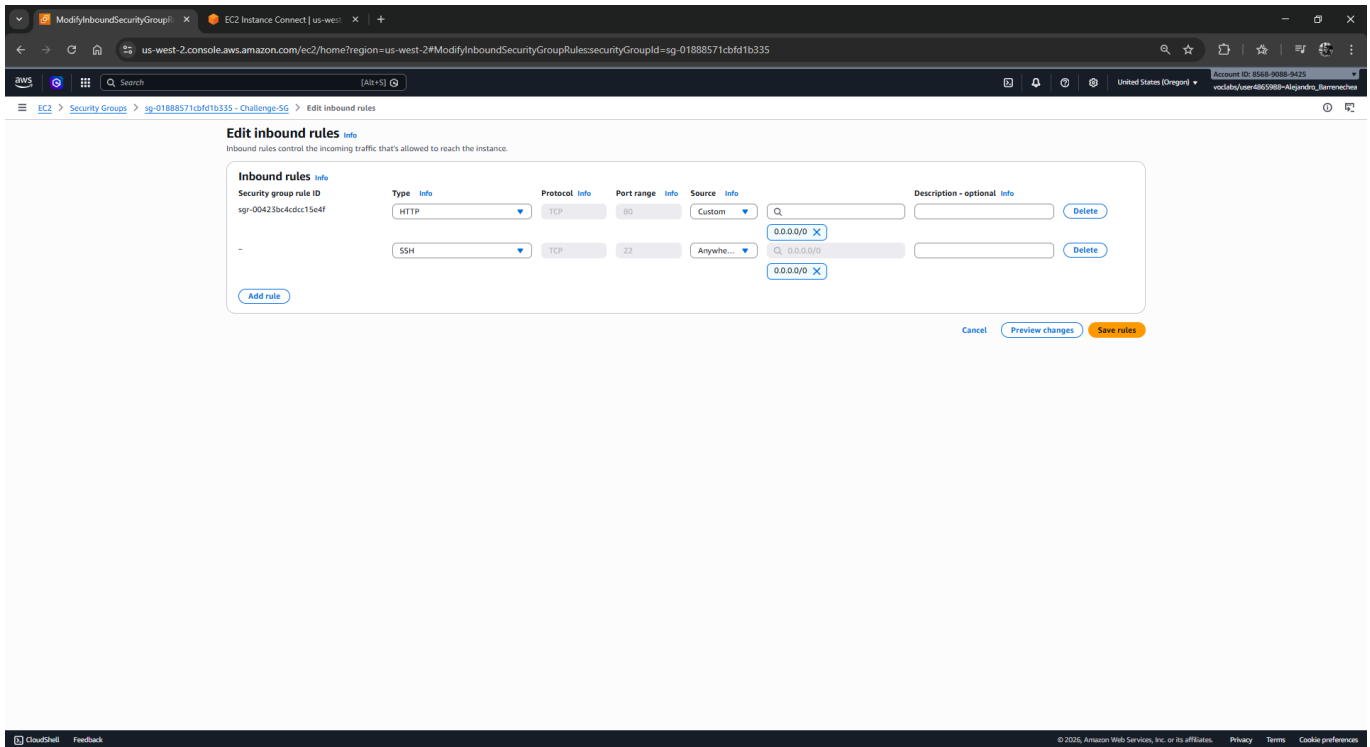
3. Solución:

- **Navega** a **Security Groups** en la consola de EC2.
- **Selecciona** el grupo asociado a esa instancia.
- **Agrega** una regla de entrada para **SSH (22)** desde **0.0.0.0/0**.

4. Resultado:

El botón "Connect" ahora establece sesión exitosamente.





5 Desafío Opcional 2: Diagnóstico de Instalación Web (HTTP)

1. **Tarea:** Acceder a la URL pública de **Misconfigured Web Server**.
2. **Fallo detectado:** El navegador no responde (Puerto 80 bloqueado o servicio apagado).

The screenshot shows the AWS Management Console for the 'us-west-2' region. The 'Instances' page displays a table of EC2 instances. The 'Misconfigured Web Server' instance is selected, and its details are shown. The instance is in a 'Running' state, using the 't2.micro' instance type, and is associated with the 'Challenge-SG' security group. The instance's public IP address is 18.236.191.26.

The screenshot shows a web browser displaying a connection error. The address bar shows the URL 'ec2-18-236-191-26.us-west-2.compute.amazonaws.com'. The page content indicates that the connection to the website was refused. The error message is in Spanish: 'No se puede acceder a este sitio web'. Below this, it states: 'La página ec2-18-236-191-26.us-west-2.compute.amazonaws.com ha rechazado la conexión.' The error code 'ERR_CONNECTION_REFUSED' is visible. There are two buttons at the bottom: 'Volver a cargar' (Reload) and 'Detalles' (Details).

3. Solución:

- **Agrega** una regla de entrada para **HTTP (80)** desde **0.0.0.0/0** en su Security Group.
- **Conéctate** a la instancia y verifica el servicio Apache:

```
sudo systemctl start httpd
sudo systemctl enable httpd
```

4. Resultado: La página web del laboratorio carga correctamente.

EC2

Dashboard

AWS Global View

Events

Instances

Instances Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#SecurityGroup:securityGroupId=sg-01888571cbfd1b335 - Challenge-SG

inbound security group rules successfully modified on security group (sg-01888571cbfd1b335) (Challenge-SG)

Details

sg-01888571cbfd1b335 - Challenge-SG

Details

Security group name

Challenge-SG

Security group ID

sg-01888571cbfd1b335

Description

Enable SSH ingress

VPC ID

vpc-01767589f6c2f5887

Owner

856890889425

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1/2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-0d61ad841b0011deb	IPv4	SSH	TCP	22	0.0.0.0/0	-
-	sg-00423bc4dc0c15e4f	IPv4	HTTP	TCP	80	0.0.0.0/0	-

Instances | EC2 | us-west-2

EC2 Instance Connect | us-west-2

Test Page for the Apache HTTPD

us-west-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-west-2&connType=standard&instanceId=i-0391719cb8433c9c5&osUser=ec2-user&sshPort=22&addressFamily=ipv4

Last login: Thu Apr 2 16:22:24 2026 from ec2-18-237-191-26.us-west-2.compute.amazonaws.com

Amazon Linux 2

AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.

https://aws.amazon.com/linux/amazon-linux-2023/

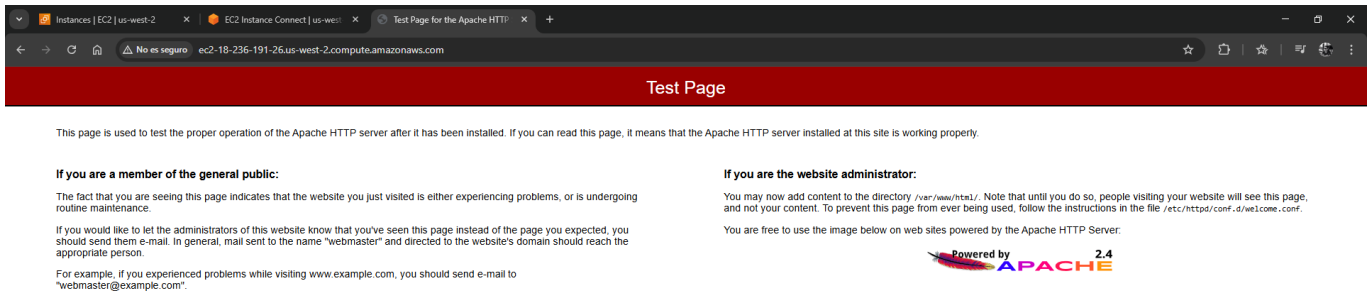
ec2-user@ip-10-0-0-89 ~\$ sudo systemctl start httpd

ec2-user@ip-10-0-0-89 ~\$ sudo systemctl enable httpd

ec2-user@ip-10-0-0-89 ~\$

i-0391719cb8433c9c5 (Misconfigured Web Server)

PublicIPs: 18.236.191.26 PrivateIPs: 10.0.0.89



🧠 5. Respuestas Analíticas

Comparativa de Métodos de Lanzamiento

- **Management Console:** Se debe usar cuando se requiere lanzar una instancia única o temporal de forma rápida y visual.
- **AWS CLI / Scripting:** Se debe usar cuando se requiere automatizar la creación de recursos de manera fiable, repetible y sin errores humanos.
- **CloudFormation:** Se debe usar para lanzar infraestructuras completas (stacks) con múltiples recursos interconectados.

Informe de Troubleshooting (Misconfigured Web Server)

1. ¿Cuál era el problema?

- Doble configuración errónea: El **Security Group** no permitía tráfico entrante por los puertos 22 (administración) ni 80 (visualización). Adicionalmente, el servicio **httpd** (Apache) no estaba en estado de ejecución.

2. ¿Qué hiciste para solucionarlo?

- Se auditaron y corrigieron las **Inbound Rules** del Security Group. Se intervino el sistema operativo mediante terminal para iniciar el servicio de servidor web manualmente.

✅ Conclusión

Has completado con éxito la gestión avanzada de instancias EC2. Has demostrado dominio en el lanzamiento manual, el uso de metadatos para automatización vía **CLI**, la implementación de **User Data** y, lo más importante, has desarrollado el criterio técnico para diagnosticar y resolver fallos de seguridad en la nube de AWS.